

30 INCHES
THE STANDARD
HEIGHT OF **MERCURY**
IN A **BAROMETER** AT
SEA LEVEL

Although the symptoms of cholera were first cataloged in the 1600s, the cholera bacterium, seen here, was not isolated until the 19th century.

THE GRAND DUKE OF TUSCANY, Italy, Ferdinand II (1610–70) invented the **sealed glass thermometer** in 1641. Working with Italian physicist Evangelista Torricelli, he improved on Galileo's thermoscope (see 1590–93) by sealing the liquid column in a glass capillary, and using wine, which did not freeze as easily as water.

A year later, Blaise Pascal invented a **mechanical**

Pascal's calculating machine

This first *Pascaline* was primarily used by accountants, and its dials were calibrated in accordance with the French currency.

calculating machine to help his father with his work in government taxes. Known as the *Pascaline*, this device worked by a system of wheels and gears and could perform routine arithmetic functions of addition, subtraction, division, and multiplication.

Dutch physician **Jacobus Bontius** (1591–1631) had traveled to the tropical East Indies for the Dutch East India Company in 1627. In 1642, his medical treatise—*De Medicina Indorum* (*Indian Medicine*) was

published posthumously. This book contained some of the **earliest descriptions of tropical diseases**, including beriberi and cholera.

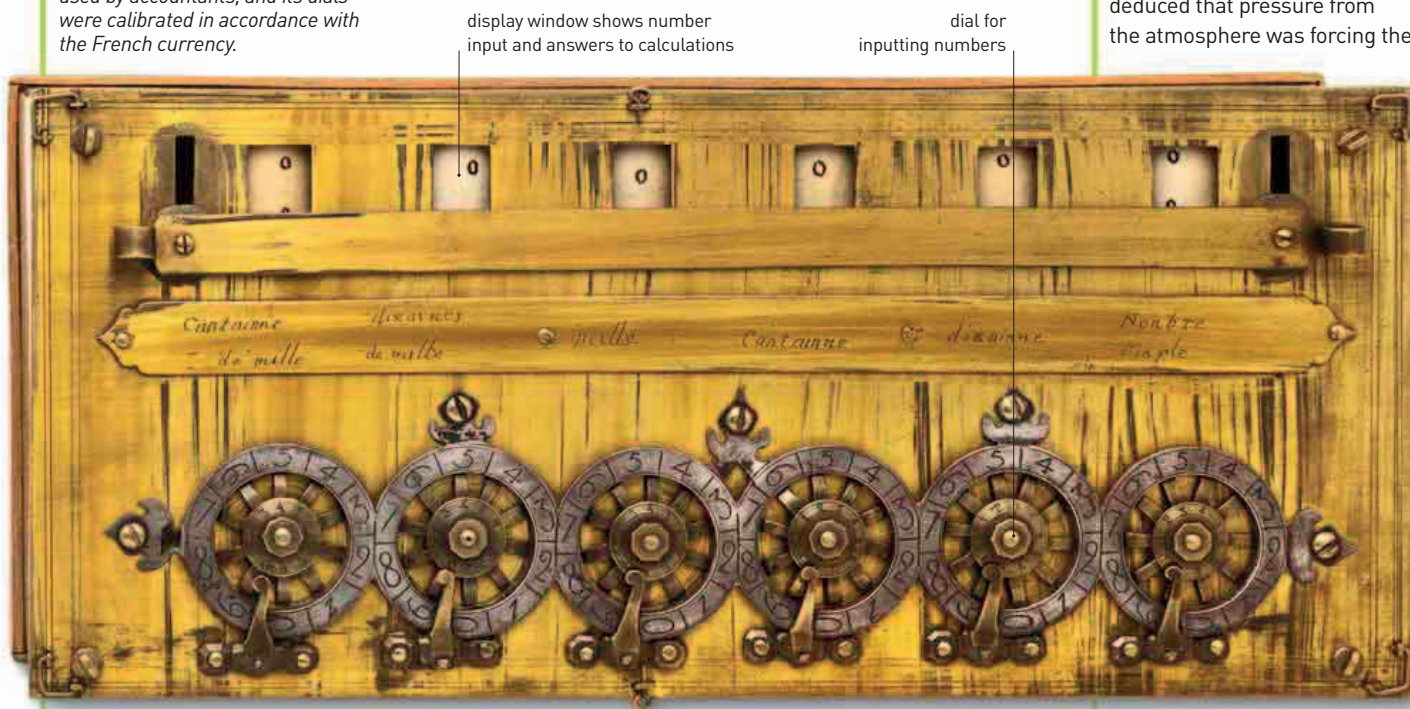
Another Dutchman, explorer and merchant **Abel Tasman** (1603–59), became the first European to reach Van Dieman's Land (now called Tasmania in his honor). He went on to visit New Zealand and islands of the Southwest Pacific. On his voyages, he recorded the earliest **European observations of Australasian fauna and flora**.

IN THE EARLY 1640S, EVANGELISTA TORRICELLI was investigating the practical problems associated with pumping water from deep wells. He imitated the action of a suction pump in a small tube and used a denser substance, mercury, instead of water, to study the effects. Torricelli discovered that mercury would rise into the sealed tube to a fixed height of 30 in (76 cm), and leave a gap at the top, which later became known as the **Torricelli vacuum**. He deduced that pressure from the atmosphere was forcing the

Torricellian tube
Evangelista Torricelli's device consisted of an evacuated glass tube containing a mercury column—the height of which was determined by atmospheric pressure.

mercury up the tube, and its height was determined by the value of this pressure (see p.106). Later, it was discovered that the pressure value varied according to altitude and weather, and small changes in atmospheric pressure signaled impending changes in weather. Torricelli's instrument therefore came to be adopted as **the first barometer**.

In 1644, René Descartes published *Principles of Philosophy*, in which he proposed an entirely **mechanical basis for the Universe**. He proposed that the Universe was filled with small, unobservable particles of matter that were set in motion by God, and that all aspects of science could ultimately be explained in accordance with this mechanical principle.



1641 Ferdinand II invents a sealed liquid thermometer

1641 Dutch scientist Franciscus Sylvius describes the deep cleft that separates lobes of the brain

March 2, 1642 German anatomist Johann Georg Wirsung discovers the pancreatic duct

1642 Blaise Pascal makes a wooden calculating machine

1642 Dutch physician Jacobus Bontius publishes the first book on tropical medicine

November 24, 1642 Dutch explorer Abel Tasman becomes the first European to see the land now called Tasmania

1643 Evangelista Torricelli develops a forerunner of the mercury barometer

1644 Italian mathematician Pietro Mengoli poses the **Basel Problem**: the precise value of a particular number series

1644 Torricelli publishes *Opera Geometrica*, and applies Galileo's laws of motion to fluids

1644 René Descartes publishes *Principles of Philosophy*, describing a mechanical universe

1644 Italian astronomer Giovanni Odierna describes the microscopic appearance of a fly's eye in the first book on microscopic life

